WEEK 11: ORGAN SYSTEM

REPRODUCTIVE SYSTEM

- Responsible for production of gametes (sperm & ova), secretion of sex hormones, and facilitation of fertilization (continuation of species)
- **Two major divisions:**
    - ***Primary (Gonadal) organs***
        - Testes and ovaries
        - <u>Function:</u> gamete production and hormone secretion
    - ***Accessory organs and structures***
        - Duct systems, glands, and external genitalia
        - <u>Function:</u> transport, nourish, and protect gametes

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MALE REPRODUCTIVE SYSTEM

- For production, maturation, storage, and delivery of spermatozoa
- Secretion of testosterone
    - For sexual reproduction, hormonal regulation, and 2nd male characteristics

PRIMARY FUNCTIONS

1.) Production of spermatozoa (spermatogenesis)

- <u>Starts production of sperm cells within the seminiferous tubules of testes</u>
- Begins in puberty and continues throughout life
    - Spermatogonia (stem cells) undergo mitosis and meiosis to for spermatids
    - Spermiogenesis transforms spermatids to spermatozoa with head, midhead (contains mitochondria for motility), and tail

2.) Delivery of sperm to female reproductive system

- Pathway:
    - Testes -> epididymis (final maturation & storage) -> vas deferens -> ejaculatory duct -> urethra
- Seminal vesicles, prostate gland, and bulbourethral gland add seminal fluid, forming semen
- Penis serves as the organ of copulation

3.) Secretion of androgens (male sex hormones)

- <u>Leydig cells produce testosterone</u> (principal androgen)
    - For 2nd sexual characteristics (deep voice, facial/body hair, muscle mass)
    - Spermatogenesis
    - Maintenance of libido
    - Regulation of gonadotrophin release through negative feedback on the hypothalamic-pituitary-gonadal (HPG) axis

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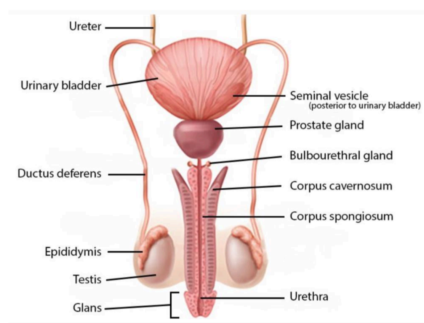
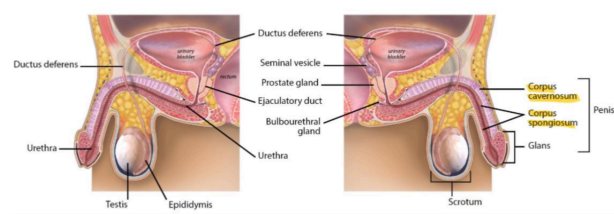
ANATOMICAL DIVISIONS





1.) Primary sex organs (gonads)

- <u>Testes</u> - produce sperm and testosterone

2.) Secondary (accessory) ducts and glands

- <u>Epididymis, vas deferens, ejaculatory ducts, and urethra</u> - transport and maturation of sperm

WEEK 11: ORGAN SYSTEM

- Seminal vesicles, prostate gland, and bulbourethral glands - secrete fluids that make up semen

3.) External genitalia

- Penis and scrotum - involved in copulation and thermoregulation of testes

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HORMONAL REGULATION (HYPOTHALAMIC-PITUITARY-GONADAL AXIS)

- Hypothalamus secretes GnRH, stimulates anterior pituitary gland to release:

- LH - stimulates Leydig cells to produce testosterone
  - Located laterally in the connective tissue of seminiferous tubules
- FSH - acts on Sertoli cells to promote spermatogenesis
  - Located medially in the connective tissue of seminiferous tubules

- Testosterone and inhibin provide negative feedback to regulate hormone levels and maintain homeostasis

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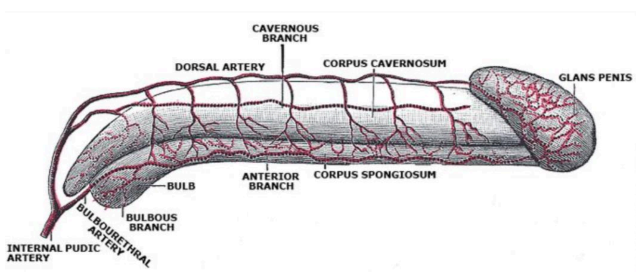
ERECTILE TISSUE AND MECHANISM OF ERECTION



- Contain numerous vascular sinusoids (cavernous spaces)

- Parasympathetic stimulation (pelvic splanchnic nerves, S2-S4) vasodilates penile arteries

- Increased blood flow fills cavernous sinusoids, while venous outflow is compressed -> erection

- Sympathetic stimulation (T11-L2) mediates ejaculation by causing contraction of smooth muscles in the reproductive ducts and glands

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SPERM STRUCTURE (SPERMATOZOON)

- Male gamete — haploid cell (23)
- Mature spermatozoon: 50-60 um

1.) Head

- Shape: Oval (flattened) — ⅓ of sperm length
- Contents:
  - Nucleus: contains haploid chromosomes
  - Acrosome: cap-like vesicle covering the anterior ⅔ of the nucleus
    - From golgi apparatus
    - Contains hydrolytic enzymes (e.g., hyaluronidase, acrosin) that digest the zona pellucida for sperm penetration
- Function: carries genetic information and enzymes

2.) Midpiece (neck region)

- Contents:
  - Centrioles - connects the nucleus to the flagellum and initiate tail formation
  - Mitochondrial sheath - surrounds the central core (axoneme) for ATP energy for motility
- Function: powerhouse of sperm, generating energy for flagellar movement

3.) Tail — flagellum

- Longest portion, responsible for motility
- Contents:
  - Principal piece - main portion of tail; contains axoneme (a 9 + 2 arrangement of microtubules) surrounded by fibrous sheath
  - End piece - terminal part; only microtubules and plasma membrane
- Distance: 1-4 mm/min through whip-like motion
- Function: Provides locomotion

WEEK 11: ORGAN SYSTEM

SPERM PATHWAY — SEVEN UP
- **S**eminiferous tubules of testes -> **E**pididymis -> **V**as Deferens -> **E**jaculatory Duct -> **N**one -> **U**rethra -> **P**enis

CLINICAL RELEVANCE
1.) Peyronie's disease
- Fibrous plaque -> abnormal curvature during erection
- Cured through shockwave therapy

2.) Balanitis
- Inflammation of glans penis

3.) Cowperitis
- Inflammation of Cowper's gland

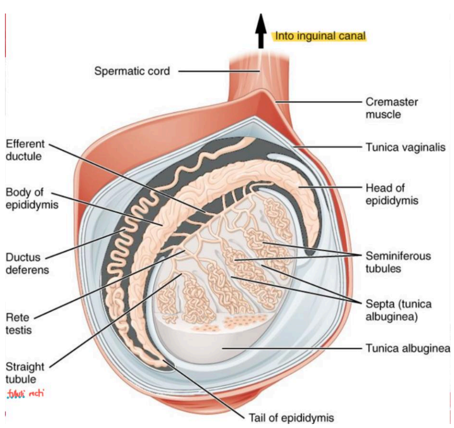
6.) Teratospermia
- Abnormally shaped sperm (e.g., double heads or tails)

5.) Asthenospermia
- Reduced sperm motility

7.) Azoospermia
- Absence of sperm in semen, indicating testicular or ductal pathology

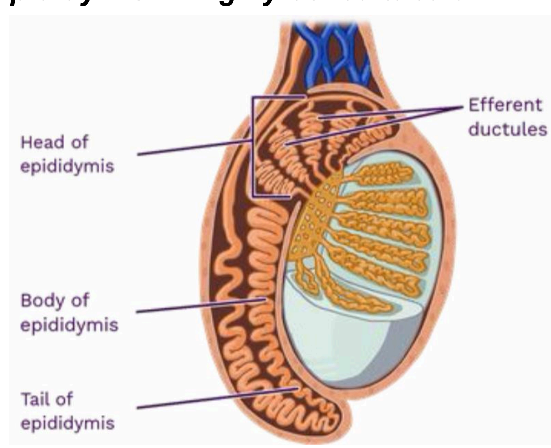
8.) Oligospermia
- Low sperm count

ANATOMICAL COMPONENTS
1.) Testes



- <u>Location:</u> within the scrotum for temperature regulation (sperm: ~2-3 °C below core body temperature)
- <u>Coverings (superficial to deep):</u>
  - Skin
  - **Dartos muscle** - smooth muscle; regulates scrotal wrinkling (temperature regulation, conservation of temperature)
  - External spermatic fascia
  - **Cremaster muscle** - skeletal muscle; elevates testes (thermal elevator)
  - Internal spermatic fascia
  - **Tunica vaginalis** - smooth movement of testis within scrotum
  - **Tunica albuginea** - dense fibrous capsule forming septa dividing the testis into lobules
- <u>Internal structure:</u>
  - ~250 lobules (1-4 seminiferous tubules each for spermatogenesis)
  - Leydig cells between tubules secrete testosterone
- <u>Blood supply:</u> testicular artery (from abdominal aorta)
- <u>Venous drainage:</u> pampiniform plexus -> testicular vein (helps cool arterial blood)
- <u>Innervation:</u> Sympathetic fibers (T10-T11)

2.) Epididymis — highly-coiled tubular



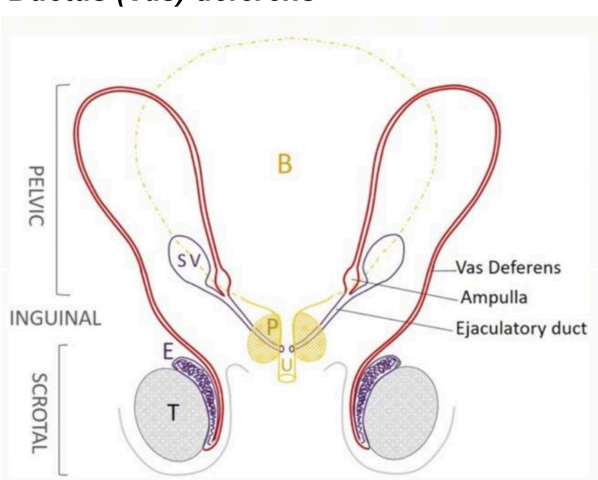
- <u>Location:</u> lies along the posterior surface of each testis
- 6 meters (20 feet) when uncoiled - longer time for maturation
- <u>Three regions:</u>

WEEK 11: ORGAN SYSTEM

- *Head (caput)* - located at the superior pole of testis; receives immature sperm from efferent ductules of testis
- *Body (corpus)* - middle portion; further sperm maturation, gains mobility and membrane stabilization
- *Tail (cauda)* - located at the inferior pole of testis; main storage site of mature sperm before ejaculation

FUNCTIONS:
1.) Sperm maturation
- Immature sperm at seminiferous tubules (non-motile and incapable of fertilization) -> pass through epididymal duct for maturation by testosterone (gain motility by flagella, capable of fertilization, can bind to zona pellucida of oocyte)
2.) Sperm storage
- The tail serves as a reservoir for mature sperm until ejaculation; if not ejaculated, it will be phagocytosed
3.) Absorption of excess testicular fluid
- Ensures efficient sperm transport and storage
4.) Secretion of nutrients and proteins
- Secrete glycoprotein, lipids, and enzymes
5.) Transport of sperm
- Through ejaculation (sympathetic stimulation), propelling sperm into the vas deferens

3.) Ductus (Vas) deferens

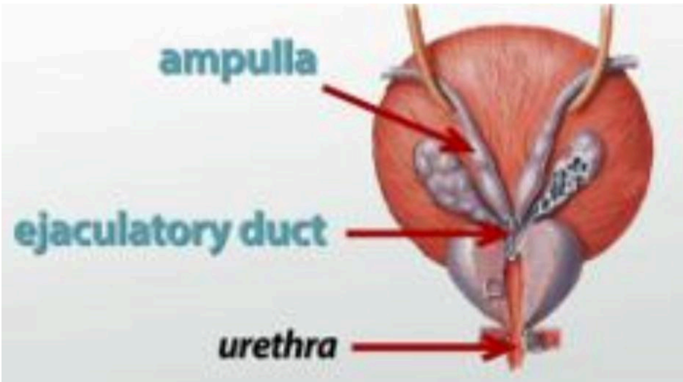


- Thick-walled, muscular tube; 45 cm (18 inches)
  - Outer and inner longitudinal, middle circular
- <u>Main conduit for sperm transport</u> from epididymis to the ejaculatory duct during ejaculation
  - Critical component of spermatic cord
- Lined with pseudostratified ciliated columnar epithelium
- <u>Anatomical course</u>
  - *Scrotal part* - **continuation of the tail of epididymis**; ascends within spermatic cord along the testicular artery, pampiniform plexus of veins, lymphatic vessels, and autonomic nerves
  - *Inguinal part* - **communication between the scrotum and pelvic cavity**; passes through the inguinal canal, entering via superficial inguinal ring and exiting through the deep inguinal ring
  - *Pelvic part* - once inside the pelvic cavity, vas deferens passes inferomedially over the ureter ("water under the bridge"), and descends posterior to the urinary bladder, where it expands to form ampulla
  - *Terminal part (ampulla and ejaculatory duct)* - dilated region that serves as a **temporary storage site of sperm**; vas deferens joins with the duct of seminal vesicle to form ejaculatory duct, which opens into prostatic urethra

FUNCTIONS:
1.) Transport of sperm
- Sperm from epididymis to ejaculatory duct during ejaculation
2.) Sperm storage
- Ampulla acts as a temporary reservoir (viable for limited time) for sperm before ejaculation
3.) Contribution to ejaculate
- Contributes a small amount of fluid (ions and proteins) to maintain sperm viability

WEEK 11: ORGAN SYSTEM

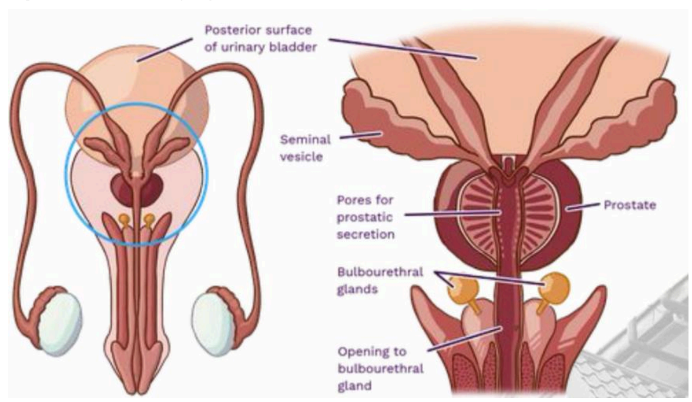
4.) Ejaculatory duct — paired short tubes



- <u>Location:</u> run obliquely through the posterior part of prostate gland to open into the prostatic urethra, specifically at the vermontanum (seminal colliculus), a small elevation on the posterior wall of urethra
- 2 cm long
- Final common pathway for the sperm and seminal vesicle secretions before entering the urethra
    - **Conduits for semen during ejaculation (sperm and seminal fluid mix properly)**
- <u>Anatomical formation and location — formed by union of 2 structures</u>
    - Duct of seminal vesicle and ampulla of vas deferens
        - *Union occurs posterior to the urinary bladder*, within prostate gland
- <u>Course and openings</u>
    - *Origin* - begins where the ampulla of vas deferens joins the duct of the seminal vesicle
    - *Pathway* - passes anteroinferiorly through the prostatic substance, running close together, one on each side of the prostatic utricle (a small pouch homologous to the female uterus and vagina)
    - *Termination* - opens into the prostatic urethra near the prostatic utricle, allowing sperm and seminal fluid to enter the urethra and mix with prostatic secretion

FUNCTIONS:
1.) Conduit of semen
- Transport semen (mixture of spermatozoa from vas deferens and seminal fluid from the seminal vesicles) into the prostatic urethra during ejaculation
2.) Integration point for secretions
- Ensures mixing of sperm with seminal vesicle fluid (rich in fructose and prostaglandins) and prostatic secretions, which together form seminal plasma (nutrient medium for sperm survival and motility)
3.) Valve mechanism
- Urination: Internal urethral sphincter prevents semen from entering the bladder
- Ejaculation: ejaculatory ducts open while the bladder neck closes

5.) Accessory glands



- Vital role in the production of seminal fluid, which nourishes, protects, and facilitates the transport of spermatozoa
- Contribute to 90% of total semen volume; 10% for sperm itself
- <u>Major accessory glands:</u>

<u>I. Seminal vesicle</u>
- Paired, elongated, sac-like glands; posterior to urinary bladder, lateral to the vas deferens ampulla
- 5 cm long; lies anterior to the rectum (palpable during digital rectal examination)
- <u>Secretion:</u> viscous, alkaline, fructose-rich secretion (60-70% of semen volume)

WEEK 11: ORGAN SYSTEM

FUNCTIONS
1.) Fructose - primary energy source for sperm motility and survival
2.) Prostaglandin - enhance smooth muscle contractions in the female reproductive tract (sperm toward oocyte)
3.) Clotting proteins (fibrinogen-like substances) - initial coagulation of semen after ejaculation. helping it adhere within the vagina
4.) Alkalinity - neutralizes acidic environment of the male urethra and female vagina

II. Prostate gland

- Single, walnut-sized gland; surrounds the prostatic urethra just below the urinary bladder
- Weighs about 20 g in adults; enclosed by a fibromuscular capsule (for strength and contraction)
- **Four zones:**
  - Peripheral zone - site of prostate cancers; largest; felt during prostate exam
  - Central zone - surrounds ejaculatory ducts
  - Transitional zone - commonly involved in **benign prostatic hyperplasia (BPH)** - urinary obstruction in older men; near bladder
  - Anterior fibromuscular stroma - mostly non-glandular tissue
- Secretions: thin, milky, slightly acidic fluid (25-30% of semen volume)

FUNCTIONS
1.) Citric acid - serves as an energy source
2.) Prostate-specific antigen (PSA) - enzyme that liquefies semen after ejaculation. releasing sperm from the coagulum
3.) Proteolytic enzymes (fibrinolysin, acid phosphatase) - enhance sperm motility by prevent clot formation

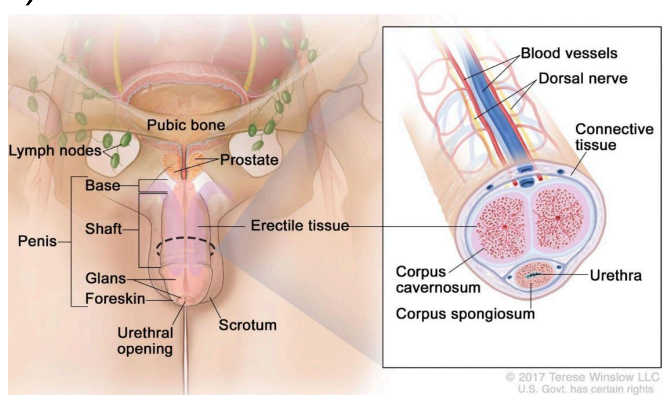
4.) Zinc and ions - stabilize sperm chromatin and aid in membrane integrity

III. Bulbourethral (Cowper's gland)

- Pair of small, pea-sized glands; located at the base of the penis, within the deep perineal pouch, on either side of the membranous urethra
- Gland's duct opens into the spongy (penile) urethra
- Secretion: clear, mucus-like, alkaline pre-ejaculate fluid (<5% of semen volume)

FUNCTIONS:
1.) Lubrication - coats and lubricates the urethra for smooth sperm passage
2.) Neutralization - neutralizes residual acidicty from urine in the urethra
3.) Pre-ejaculatory emission - secreted during sexual arousal, before ejaculation

6.) Penis



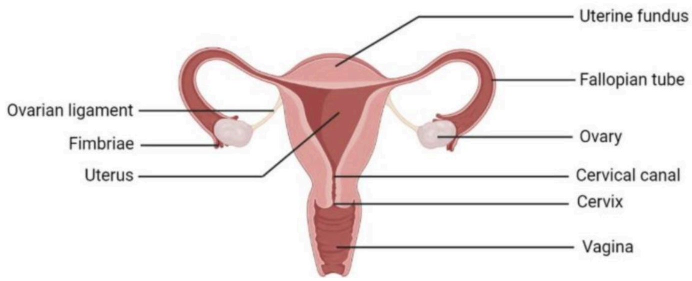
- Anatomical division:

- Root
  - Anchors the penis to the pubic arch and perineal membrane
  - Composed of **bulb** (central part of corpus spongiosum) and **crura** (proximal ends of the corpora cavernosa)

  - Secured by **bulbospongiosus and ischiocavernosus** for erection and ejaculation
- <u>Body (shaft)</u>
  - Free, pendulous portion of the penis extending from the root to the glands
  - Enclosed by skin and fascia, containing 3 cylindrical masses of erectile tissue:
    - Two corpora cavernosa (dorsal) - responsible for penile rigidity during erection; primary penile structure for erection
    - One corpus spongiosum (ventral) - surrounds the spongy (penile) urethra and prevents its collapse during erection, allowing for semen and urine passage
- <u>Glans penis</u>
  - Expanded distal end of the corpus spongiosum
  - Covered by prepuce (foreskin), unless circumcised
  - Contains high concentration of sensory nerve endings, making it extremely sensitive and important for sexual stimulation

WEEK 11: ORGAN SYSTEM

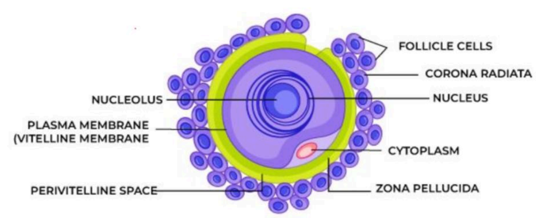
FEMALE REPRODUCTIVE SYSTEM
- For production of ova, reception of sperm, fertilization, development of embryo and fetus, and secretion of reproductive hormones



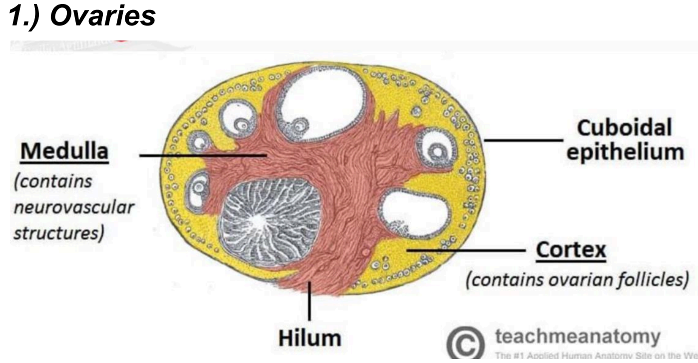
MAJOR FUNCTIONS
1.) Production of ova
- From <u>ovaries</u> through oogenesis
- Develops a follicle which undergoes changes during menstrual cycle
2.) Reception of sperm
- <u>Vagina</u> acts as the receptacle for sperm
- <u>Cervix</u> and uterus assist in transporting sperm towards the ampulla of fallopian tube
3.) Site for fertilization and embryonic development
- Occurs in the <u>ampulla of the uterine (fallopian) tube</u>
- Uterus provides the site of implantation, nourishment, and development of the embryo and fetus
4.) Nourishment of the developing fetus
- <u>Placenta</u> in the uterus facilitates exchange of gases, nutrients, and wastes between the mother and fetus
- <u>Mammary glands</u> provide nutrition through lactation
5.) Hormone secretion
- Ovaries secrete estrogen and progesterone
  - Regulates menstrual cycle
  - Maintain 2nd sexual characteristics
  - Support pregnancy and mammary gland

THE OVUM (HUMAN EGG CELL)
- Female gamete — haploid cell (23)
- Produced by ovary during oogenesis
- One of the largest cells in the body
  - 120 um in diameter

1.) Corona radiata
- <u>Location:</u> outermost
- <u>Function:</u> nourishment, protection, sperm guidance; metabolic support
2.) Zona pellucida
- <u>Location:</u> glycoprotein layer beneath corona
- <u>Function:</u> sperm binding, species specificity, block to polyspermy
3.) Pervitelline space
- <u>Location:</u> between zona and membrane
- <u>Function:</u> contains polar body, sperm fusion site
4.) Vitelline membrane (oolema)
- Location: plasma membrane of ovum
- Function: entry; fuses with sperm membrane
5.) Ooplasm (cytoplasm)
- Location: internal matrix
- Function: nutrient storage, organelle function, metabolism
6.) Nucleus (female pronucleus)
- Location: central or eccentric
- Function: genetic material
7.) Polar bodies
- Location: in perivitelline space
- Function: discard excess chromosomes during meiosis



ANATOMICAL COMPONENTS
1.) Ovaries



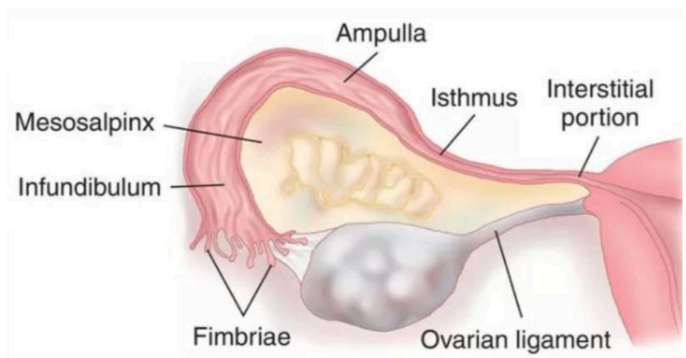
WEEK 11: ORGAN SYSTEM

- Location: lateral walls of the pelvic cavity, on each side of the uterus
- 1 cm long, 1.5 cm wide, and 1 cm thick
- Structures:
  - **Cortex**
    - Outer dense region; contains ovarian follicle in primordial, primary, secondary, and mature/Graafian follicles
    - Contains developing oocyte surrounded by follicular (granulosa) and thecal cells
    - After ovulation, follicle -> corpus luteum (secretes progesterone)
    - If no fertilization, it degenerates into corpus albicans (fibrous scar)
  - **Medulla**
    - Inner vascular region; for supplying nutrients and hormonal support to the ovarian tissue
- Held in position by:
  - **Ovarian ligament** - from the ovary to the lateral side of uterus
  - **Suspensory ligament** - from ovary to the lateral pelvic wall; carries ovarian artery, vein, lymphatics, and nerves
  - **Mesovarium** - ovary to the broad ligament of the uterus

FUNCTIONS
1.) Oogenesis
- Immature ova mature and are released during ovarian cycle (about every 28 days; one ova per cycle through ovulation)
2.) Hormone secretion
- Estrogen
  - Produced by developing follicles and corpus luteum
    - For growth and proliferation of endometrium during the proliferative phase of the menstrual cycle
    - Development of 2nd sexual characteristics
    - Maintains bone density and cardiovascular health

- Progesterone
  - Produced by corpus luteum after ovulation
    - Prepares endometrium for implantation of fertilized ovum
    - Inhibits uterine contraction during pregnancy
    - Development of mammary glands
    - Maintains uterine lining throughout pregnancy until placental hormone takes over

2.) Uterine (Fallopian) tubes



- Location: lies within the superior margin of the broad ligament of the uterus
- 10-12 cm pair of muscular tubes; passageway for ova from ovaries to uterus and the ampulla
- Lined with ciliated epithelium
  - Cilia + smooth muscle = contraction

PARTS & FUNCTION
1.) Fimbriae
- Finger-like projections extending from the margin of infundibulum; fimbria ovarica (largest) is attached directly to ovary
- Function: Sweeps ovulated oocyte from the ovarian surface -> infundibulum during ovulation
2.) Infundibulum
- Funnel-shaped lateral end of the uterine tube, position very close to the ovary; opens into the peritoneal cavity and is bordered by fimbriae
- Function: receives oocyte from fimbria -> uterine tube

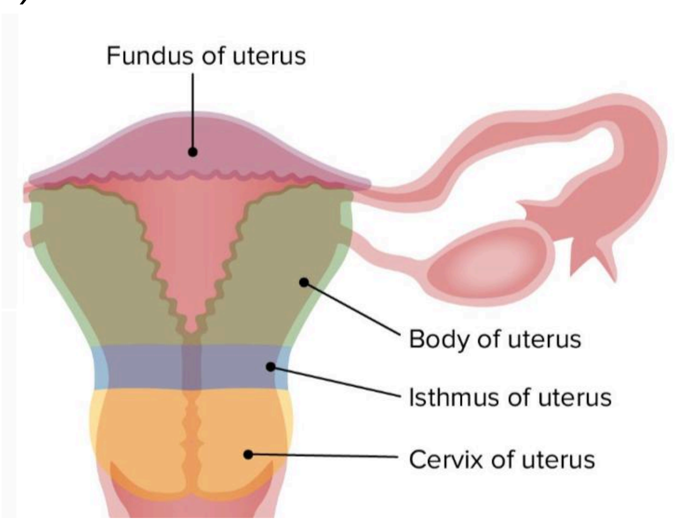
WEEK 11: ORGAN SYSTEM

3.) Ampulla
- Widest and longest segment (⅔); lined with ciliated and secretory epithelial cells (aid in oocyte of sperm transport)
- Function: primary site of fertilization

4.) Isthmus — common site of ectopic pregnancy
- Narrow, thick-walled medial portion connecting ampulla -> uterus
- Function: sphincter-like segment for the passage of zygote -> uterus; performs peristaltic contraction

5.) Intramural (interstitial) part
- Shortest portion which passes through the uterine wall and opens into uterine cavity at uterine horn
- Function: entrance of the uterine tube -> uterus (allows zygote to reach ampulla)

3.) Uterus



- Location: pelvic cavity, between bladder and rectum
- Parts:
    - *Fundus* - superior rounded region
    - *Body* - main portion
    - *Cervix* - narrow inferior portion; opens into vagina
- Layers:
    - Perimetrium - outer serous coat
    - Myometrium - thick smooth muscle layer (contracts during childbirth)
    - Endometrium - inner mucosa; undergoes changes during menstrual cycle

FUNCTIONS
1.) Site of implantation, embryonic, and fetal development
2.) Undergoes cyclic endometrial changes in response to ovarian hormones
3.) Contracts during parturition to expel fetus
4.) Provides nutritional support to the developing fetus through placenta

4.) Vagina
- Locations:
    - Anteriorly - posterior to the urinary bladder and urethra
    - Posteriorly - anterior to the rectum
    - Superior end - surrounds the cervix forming fornices (anterior, posterior, and lateral)
    - Inferior end - opens externally into the vaginal vestibule

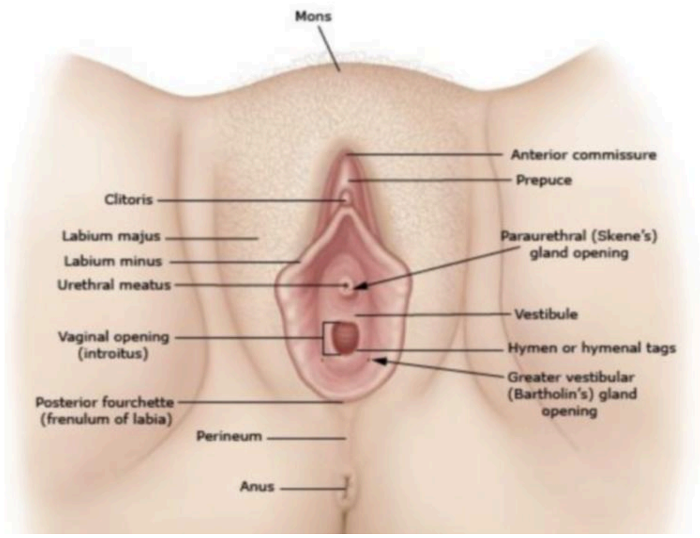
WALL LAYERS
1.) Mucosa
- Lined with stratified squamous epithelium for resisting friction
- Lacks glands; lubrication comes from *cervical secretions and vestibular glands*
- Contains rugae for stretching during childbirth and intercourse
- Maintains acidic pH (due to lactic acid produced by resident bacteria) for protection

2.) Muscularis
- Composed of smooth muscle fibers (inner circular and outer longitudinal layers)
- Allows distensibility during intercourse and childbirth

3.) Adventitia
- Outer fibrous connective tissue layer that anchors the vagina to surrounding structures

5.) External female genitalia (vulva)

WEEK 11: ORGAN SYSTEM



- Also known as **pudendum**
- Protects the internal reproductive organs, provides sensory pleasure (arousal), and facilitate the passage of urine and childbirth

PARTS & FUNCTIONS

1.) Mons pubis
- Rounded, fatty eminence anterior to the pubic symphysis; *covered with pubic hair after puberty* (escutcheon)
- Function: cushions and protects the pubic bone during intercourse

2.) Labia majora
- *2 large, longitudinal folds of skin* extending from mons pubis to the perineum; contain *sebaceous and sweat glands, and subcutaneous fat*
- Function: enclose and protect the other external genital organs; **homologous to scrotum**

3.) Labia minora
- *Thin, hairless folds of skin* medial to the labia majora; enclose the vestibule; vascular with nerve endings
- Function: protect vaginal and urethral opening; form part of the clitoral hood anteriorly

4.) Clitoris
- *Small, cylindrical erectile structure* anteriorly at the junction of the labia majora; composed of corpora carvenosa and glans clitoris
- Function: highly sensitive; primary organ of female sexual arousal; **homologous to penis**

5.) Vestibule
- Cleft between the labia minora that contains *openings of the urethra, vagina, and vestibular gland ducts*
- Function: entrance to both urinary and reproductive

6.) External urethral orifice
- Between clitoris and vaginal opening
- Function: passage of urine from the urethra to the exterior

7.) Vaginal orifice (introitus)
- Opening of the vagina into the vestibule; *partially covered by the hymen in virgins*
- Function: entry for penis during intercourse; exit for menstrual blood and childbirth

8.) Greater vestibular (Bartholin's) glands
- Located on either side of the vaginal orifice; opens into vestibule
- Function: *secrete mucus* for vaginal lubrication during sexual arousal

9.) Lesser vestibular (Skene's) glands
- Located near the urethral orifice
- Function: *secrete mucus* for vestibular and urethral opening lubrication

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STAGES OF SEXUAL AROUSAL — HUMAN SEXUAL RESPONSE CYCLE

1.) Excitement phase — arousal phase
- Initial stage of sexual stimulation
- Nervous control:
  - **Parasympathetic fibers (S2-S4)** -> vasodilation and secretory activity
  - Neurotransmitter involved: **NO** -> smooth muscle relaxation -> increased blood flow

MALE
- Penile erection due to vasodilation of penile arteries and filling of corpora cavernosum
- Testes elevate and enlarge within scrotum

FEMALE
- Clitoral erection; vaginal lubrication from Bartholin's glands; swelling of labia and breasts
- Nipples become erect; uterus elevates slightly

COMMON RESPONSES
- Increased HR and BP, muscle tension, skin flushing
- physiological arousal, anticipation, and

WEEK 11: ORGAN SYSTEM

increased sensitivity to touch

2.) Plateau phase

- Intensification and maintenance of physiological changes; preparation for orgasm
- Nervous control:
  - **Parasympathetic fibers (S2-S4)** -> maintains vasocongestion
  - Increased sympathetic tone; starts preparing for orgasm

MALE - Maximum penile erection; pre-ejaculate secretion from bulbourethral gland lubricates urethra - Testes drawn tightly against perineum
FEMALE - Vaginal walls engorge; clitoris retracts under prepuce; outer third of vagina forms the organismic platform - Breasts enlarge further; areolar darkens
COMMON RESPONSES - Further increased HR, RR, and muscle tension; involuntary pelvic muscle contractions may begin - Heightened sexual pleasure and tension

3.) Orgasm phase — climax

- Peak of sexual pleasure; rhythmic contraction and intense sensory experience
- Nervous control:
  - **Sympathetic fibers (L1-L2)** -> trigger emission and ejaculation
  - **Somatic pudendal nerves** -> coordinate rhythmic muscle contractions in the perineum

MALE - Ejaculation: emission (sperm and glandular secretions mix to form semen) & expulsion - Controlled by sympathetic and somatic pudendal nerve activity
FEMALE - Rhythmic contractions of uterus, vagina, and pelvic muscles (~0.8-second intervals) - Often accompanied by uterine contractions aiding sperm transport
COMMON RESPONSES - Sudden release of sexual tension; involuntary body movements, vocalization, and possible loss of awareness of surroundings - Intense pleasure, muscle spasm, and heightened sensory feedback

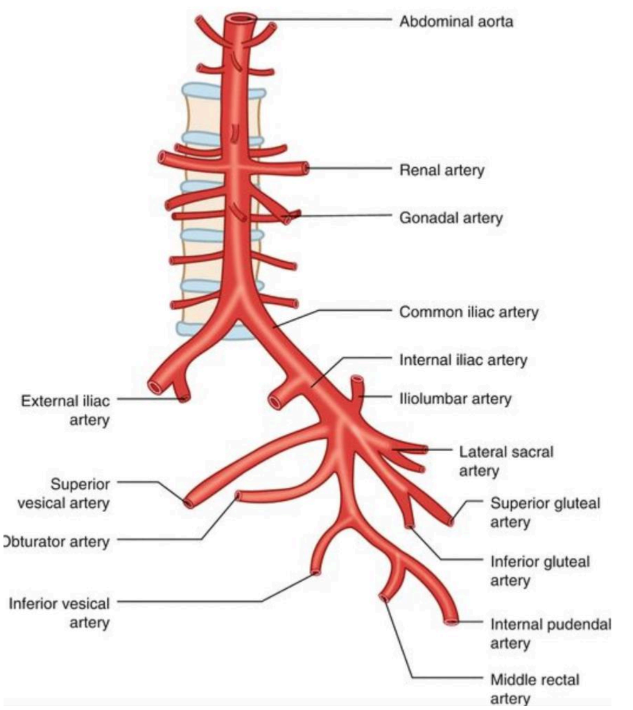
4.) Resolution phase

- Return of the body to an unaroused, resting state
- Nervous control:
  - Dominated by **sympathetic vasoconstriction** leading to detumescence (loss of erection)
  - **Prolactin and oxytocin** release promote relaxation and emotional bonding

MALE - Penis becomes flaccid as blood drains from erectile tissue; refractory period begins (duration increases with age) - Testes descend to normal position
FEMALE - Vaginal, uterine, and clitoral engorgement subsides; no refractory period (females may experience multiple orgasms - Labia and breasts return to normal size
COMMON RESPONSE - HR, BP, and RR gradually return to baseline - General sense of relaxation, satisfaction, and fatigue

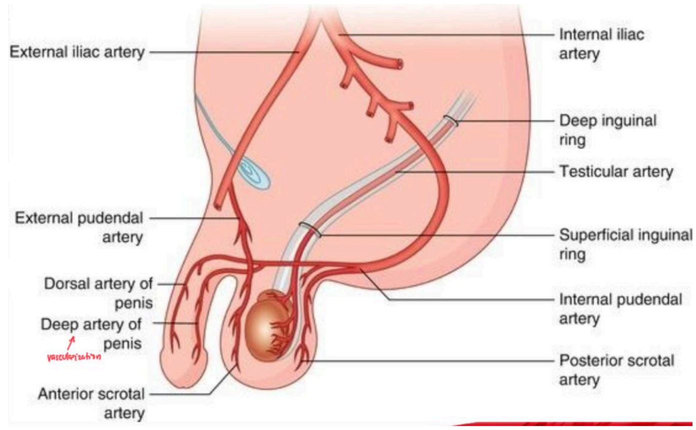
WEEK 11: ORGAN SYSTEM

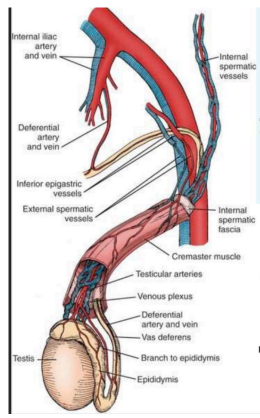
BLOOD SUPPLY FOR MALE

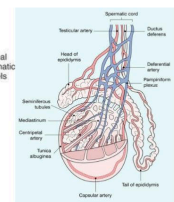


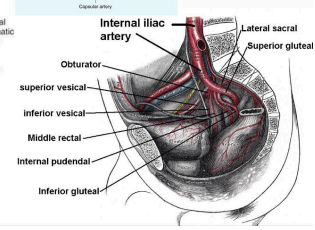
Organ	Main Artery	Origin / Arterial Pathway (Tracing from Aorta)	Notes / Distribution
Testes	**Testicular artery (gonadal artery)**	**Abdominal aorta → Testicular artery** (arises just below the renal arteries) → passes **retroperitoneally**, crosses **ureter**, enters **inguinal canal** via **deep inguinal ring**, and descends into **spermatic cord** to reach testes.	Supplies **testis** and **epididymis**. Travels with **pampiniform plexus** (venous counterpart).
Vas deferens (ductus deferens)	**Artery to the vas deferens (deferential artery)**	**Inferior vesical artery → Deferential artery → joins testicular artery** near epididymis.	Supplies **ductus deferens**, **epididymis**, and **seminal vesicle**.
Seminal vesicles	**Vesicular branches of inferior vesical artery**	**Internal iliac artery → Anterior division → Inferior vesical artery → Vesicular branches.**	Supplies **seminal vesicle** and part of the **bladder base**.
Prostate gland	**Inferior vesical artery** (also middle rectal, internal pudendal)	**Internal iliac artery → Inferior vesical artery → Prostatic branches.**	Rich vascular supply for **prostate and bladder neck**.
Penis	**Internal pudendal artery**	**Internal iliac artery → Anterior division → Internal pudendal artery → passes through pudendal canal → divides into: • Dorsal arteries of penis • Deep arteries of penis (cavernosal) • Bulbourethral artery**	Supplies **erectile tissues** (corpora cavernosa and spongiosum), **urethra**, and **skin of penis**.

Structure	Arterial Supply	Venous Drainage	Innervation
Testes	**Testicular artery** (from abdominal aorta)	**Pampiniform plexus** → testicular vein → IVC (right) / left renal vein (left)	Sympathetic fibers (T10–T11) via testicular plexus – regulate vasomotor tone and spermatogenesis
Epididymis & Vas Deferens	**Deferential artery** (from inferior vesical artery)	**Pampiniform and vesical plexuses**	Sympathetic fibers (T10–L2) – peristaltic contractions during ejaculation
Seminal Vesicles	**Inferior vesical and middle rectal arteries**	**Vesical and prostatic venous plexuses**	Sympathetic (L1–L2) and Parasympathetic (S2–S4) fibers
Prostate Gland	**Inferior vesical artery** (branch of internal iliac)	**Prostatic venous plexus** → internal iliac vein	Inferior hypogastric plexus (mixed sympathetic & parasympathetic fibers)
Penis	**Internal pudendal artery** → deep and dorsal arteries of penis	**Deep dorsal vein** → prostatic venous plexus → internal iliac vein	Pudendal nerve (somatic motor/sensory); Cavernous nerves (S2–S4) – erection (parasympathetic); Sympathetic fibers – ejaculation









BLOOD SUPPLY FOR FEMALE

Organ	Main Artery	Origin / Arterial Pathway (Tracing from Aorta)	Notes / Distribution
Ovaries	**Ovarian artery (gonadal artery)**	**Abdominal aorta → Ovarian artery** (arises below renal arteries) → travels in **suspensory ligament of ovary** → enters **mesovarium** → branches to **ovary** and **uterine tube**.	Supplies **ovary** and **lateral part of uterine tube**. Anastomoses with **uterine artery**.
Uterine tubes (fallopian tubes)	**Uterine and ovarian arteries (anastomose)**	**Internal iliac → Uterine artery → tubal branches + Ovarian artery → tubal branches** → anastomose at mesosalpinx.	Ensures **dual supply** to tubes and **ovary-fimbrial region**.
Uterus	**Uterine artery**	**Internal iliac artery → Anterior division → Uterine artery →** ascends along **lateral uterus** in the **broad ligament**, giving off **arcuate**, **radial**, and **spiral arteries** to the **myometrium** and **endometrium**.	Highly vascular for **implantation** and **pregnancy maintenance**. Anastomoses with **ovarian** and **vaginal arteries**.
Vagina	**Vaginal artery** (and uterine branches)	**Internal iliac artery → Vaginal artery** (often from **uterine** or **inferior vesical** artery).	Supplies **vaginal wall** and **cervix**. Anastomoses with **uterine and internal pudendal arteries**.
Vulva (external genitalia)	**Internal pudendal artery**	**Internal iliac artery → Internal pudendal artery → perineal and labial branches.**	Supplies **labia majora, labia minora, clitoris**, and **vestibule**.
Clitoris	**Deep and dorsal arteries of the clitoris**	**Internal pudendal artery → terminal branches (deep & dorsal arteries).**	Homologous to **penile arteries** in males; supply **erectile tissue of clitoris**.

WEEK 11: ORGAN SYSTEM

Structure	Arterial Supply	Venous Drainage	Innervation
Ovaries	**Ovarian artery** (from abdominal aorta)	**Ovarian vein** → IVC (right) / left renal vein (left)	Ovarian plexus (sympathetic & parasympathetic from T10–L1)
Uterine Tubes (Fallopian Tubes)	**Uterine and ovarian arteries (anastomosis)**	**Uterine and ovarian veins**	Pelvic splanchnic nerves (S2–S4) – peristalsis and ciliary movement
Uterus	**Uterine artery** (branch of internal iliac)	**Uterine venous plexus** → internal iliac veins	Pelvic splanchnic nerves (S2–S4) – uterine contractions (parasympathetic); Sympathetic (T12–L2) – vasoconstriction and pain pathways
Vagina	**Vaginal artery** (from internal iliac) & **uterine artery branches**	**Vaginal venous plexus** → internal iliac vein	Deep perineal branch of pudendal nerve (somatic); Pelvic splanchnic (S2–S4) for autonomic and sensory control
Vulva (External Genitalia)	**Internal pudendal artery** → perineal & dorsal arteries of clitoris	**Internal pudendal vein**	Pudendal nerve (motor/sensory); Ilioinguinal nerve (mons pubis, labia majora); Genitofemoral nerve (femoral branch) – sensory to mons pubis